Physics 4B Spring 2026
Week 6 Discussion

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1. **Brainwarmer**[1]. A balloon of mass $M = 4$ kg has a volume $V = 18$ m^3 and is filled with $m = 3.2$ kg of helium. Assuming $\rho_{\text{air}} = 1.3$ kg/m^3, compute

   (i) the buoyancy force

   (ii) the acceleration of the balloon

   Solution:

   (i) The buoyancy force is

   $$F_B = \rho_{\text{air}} V_{\text{balloon}} g = 229 \text{ N} \tag{1}$$

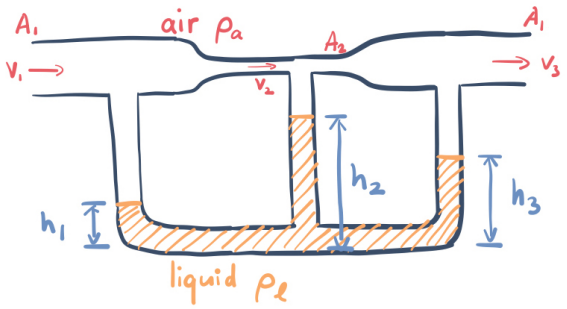
   (ii) The net force on the balloon is

   $$F_{\text{net}} = F_B - (M + m)g \tag{2}$$

   The acceleration is

   $$a = \frac{F_{\text{net}}}{M + m} = \frac{F_B}{M + m} - g = 22 \text{ m/s}^2 \tag{3}$$

2. **Venturi meter**[2]. Below is a figure showing the Venturi meter demo from class. The tube on the top has cross-sectional area of A_1 on both ends, but has a narrow segment in the middle with cross-sectional area $A_2 < A_1$.



   (a) Assuming the density of air is the same everywhere, find the middle height h_2 in terms of $\rho_a, \rho_l, A_1, A_2, v_1, h_1$.

[1] Shankar 10.11

[2] Adapted from additional homework problem

(b) Find h_3 in terms of $\rho_a, \rho_l, A_1, A_2, v_1, h_1$. What do you find?

(c) If there is friction in the top tube, how should we modify the equation we wrote? Does that explain the discrepancy you noticed?

Solution:

(a) First let's call the pressure at the bottom of the liquid P_0, the pressure at the left P_1, the pressure at the narrow segment P_2, and the pressure at the right P_3.

We want to find h_2. Since the liquid is incompressible, we can equate the pressure at the bottom of the two vertical tubes

$$P_0 = P_1 + \rho_a g(h_2 - h_1) = P_2 + \rho_l g(h_2 - h_1) \tag{4}$$

Note that P_1 and P_2 are not independent and can be related via Bernoulli's law

$$P_1 + \frac{1}{2}\rho_a v_1^2 = P_2 + \frac{1}{2}\rho_a v_2^2 \tag{5}$$

Here the speeds are also not independent and can be related via the continuity equation

$$A_1 v_1 = A_2 v_2 \tag{6}$$

Combining (4) and (5), we get

$$\rho_a g(h_2 - h_1) - \frac{1}{2}\rho_a v_1^2 = \rho_l g(h_2 - h_1) - \frac{1}{2}\rho_a v_2^2 \tag{7}$$

so the height is

$$h_2 = h_1 + \frac{\rho_a(v_2^2 - v_1^2)}{2(\rho_l - \rho_a)g} = h_1 + \frac{\rho_a v_1^2(A_1^2/A_2^2 - 1)}{2(\rho_l - \rho_a)g} \tag{8}$$

which is clearly larger than h_1 since the second term positive.

(b) We can just repeat the steps we did in the previous part to write

$$P_0 = P_1 + \rho_a g(h_3 - h_1) = P_3 + \rho_l g(h_3 - h_1) \tag{9}$$

$$P_1 + \frac{1}{2}\rho_a v_1^2 = P_3 + \frac{1}{2}\rho_a v_3^2 \tag{10}$$

$$A_1 v_1 = A_1 v_3 \tag{11}$$

The last two equations tell us that $v_1 = v_3$ and $P_1 = P_3$, so we end up with $\rho_a(h_3 - h_1) = \rho_l(h_3 - h_1)$. Since $\rho_a \neq \rho_l \neq 0$, the only possible solution is $h_3 = h_1$.

(c) If there's friction, we add a friction contribution $\mathcal{F} < 0$ to the left hand side of the Bernoulli equation (10). $\mathcal{F}$ is negative since friction does negative work on the system.

$$P_1 + \frac{1}{2}\rho_a v_1^2 + \mathcal{F} = P_3 + \frac{1}{2}\rho_a v_3^2 \tag{12}$$

Combined with (6), this implies $P_3 = P_1 + \mathcal{F} < P_1$. This results in

$$h_3 = h_1 + \frac{\mathcal{F}}{(\rho_a - \rho_l)g} > h_1 \tag{13}$$

Which is exactly what we see in the demo.

3. **Spinning bucket**[3]. Consider water spinning in a bucket with angular velocity ω. Let us prove that the surface will assume a parabolic shape. Go to the rotating frame, where in addition to gravity there is a radially outward centrifugal force, given by $\vec{F}_{\text{cf}} = m\omega^2 r\ \hat{r}$. Find the direction of the vector sum of that force and the force of gravity. This sum will be the "local force of gravity," and the surface of water will be perpendicular to it. From the tangent of the angle the vector sum makes with the x-axis, infer that $\frac{dy}{dx} = Cx$, where C is a constant. (i) What is C? (ii) Integrate this relation to show that $y(x)$ is parabolic. (iii) *Challenge*: Let the depth from the surface $y(x)$ be d, what is the pressure inside the bucket as a function of d and x?

Solution: The forces felt by a small chunk of water of mass m are

$$\vec{F}_{\text{g}} = -mg\ \hat{z} \tag{14}$$
$$\vec{F}_{\text{cf}} = m\omega^2 r\ \hat{r} \tag{15}$$

Given that the water must have cylindrical symmetry, we can consider a point along the x-axis and replace r with x. The effective force on the water would be

$$\vec{F}_{\text{eff}} = m\omega^2 x\ \hat{x} - mg\ \hat{z} \tag{16}$$

This points in the direction that makes an angle of $\theta = -\arctan\left(\frac{g}{\omega^2 x}\right)$ with the horizontal. Therefore, the tangent surface has slope

$$\frac{dy}{dx} = \frac{\omega^2 x}{g} \tag{17}$$

So $C = \omega^2 x/g$. Integrate this, we get $y(x) = \frac{\omega^2}{2g}x^2 + A$.

[3] Shankar 10.16

Now, to find the pressure as a function of x and d, we start with the hydrostatic equilibrium equation, but this time in 2d:

$$\vec{\nabla} P = \hat{x}\frac{\partial P}{\partial x} + \hat{z}\frac{\partial P}{\partial z} = \rho(\omega^2 x\ \hat{x} - g\ \hat{z}) \tag{18}$$

so we get

$$\frac{\partial P}{\partial x} = \rho\omega^2 x,\ \frac{\partial P}{\partial z} = -\rho g \tag{19}$$

The solution is

$$P(x,z) = \rho\left(\frac{1}{2}\omega^2 x^2 - gz\right) + C \tag{20}$$

Recall from the last part we found the surface is $z_s(x) = \frac{\omega^2 x^2}{2g} + z_0$. By definition, $d(x,z) = z_s(x) - z$, so

$$\begin{aligned} P(x,z) &= \rho\left(\frac{1}{2}\omega^2 x^2 - gz\right) + C \\ &= \rho\left(\frac{1}{2}\omega^2 x^2 - gz_s(x) + gd(x,z)\right) + C \\ &= \rho\left(gd(x,z) - gz_0\right) + C \end{aligned} \tag{21}$$

At $x = 0$ and $d = 0$, $P = P_{\text{atm}}$, so $C - \rho g z_0 = P_{\text{atm}}$. Therefore,

$$P(x,z) = P_{\text{atm}} + \rho g d(x) \tag{22}$$

4. **Water + ice**.

   (a) I mix 500 g of water ($c_{\text{w}} = 1$ cal/(g $\cdot$ °C)) at 20°C with 200 g of ice ($c_{\text{i}} = 0.5$ cal/(g $\cdot$ °C))at 0°C. The latent heat of fusion for water is $L = 80$ cal/g. What is the end product and how much of each component is there?[4]

   (b) If 400 g of ice at -2°C is placed in 1 kg of water at 21°C, what is the end product when equilibrium is reached?[5]

   Solution:

   (a) We need to consider 3 scenarios:

      i. The end product is water.

      ii. The end product is a mixture of water and ice.

      iii. The end product is ice (which is actually not possible here since the ice starts at 0°C and cannot get to a lower temperature).

[4] Shankar 11.3

[5] Shankar 11.6

The first scenario seems more likely since there's more water and the ice is already at the melting point, so we will consider it first. Let ΔQ_i be the heat absorbed by the object i, then we have

$$\Delta Q_{\mathrm{w}} + \Delta Q_{\mathrm{i}} = m_{\mathrm{w}} c_{\mathrm{w}} (T_f - T_{\mathrm{w},i}) + m_{\mathrm{i}} L + m_{\mathrm{i}} c_{\mathrm{i}} (T_f - T_{\mathrm{i},i}) = 0 \tag{23}$$

The final temperature is

$$T_f = \frac{m_{\mathrm{w}} c_{\mathrm{w}} T_{\mathrm{w},i} + m_{\mathrm{i}} c_{\mathrm{i}} T_{\mathrm{i},i} - m_{\mathrm{i}} L}{m_{\mathrm{w}} c_{\mathrm{w}} + m_{\mathrm{i}} c_{\mathrm{i}}} = -10\ ^\circ\mathrm{C} \tag{24}$$

which contradicts with our assumption, so we will move on to the next scenario, i.e. a mixture of water and ice. In this case, $T_f = 0^\circ$C, and we have

$$\Delta Q_{\mathrm{w}} + \Delta Q_{\mathrm{i}} = m_{\mathrm{w}} c_{\mathrm{w}} (T_f - T_{\mathrm{w},i}) + m_{\mathrm{i,melted}} L = 0 \tag{25}$$

The amount of heat the water gives off to melt the ice is absorbed by the part of ice that actually melted, so

$$m_{\mathrm{i,melted}} = \frac{-\Delta Q_{\mathrm{w}}}{L} = \frac{m_{\mathrm{w}} c_{\mathrm{w}} (T_{\mathrm{w},i} - T_f)}{L} = \frac{10000\ \mathrm{cal}}{80\ \mathrm{cal/g}} = 125\ \mathrm{g} \tag{26}$$

There's 625 g water and 75 g ice remaining.

(b) Again, we consider the 3 scenarios. It looks likely this will end up with a mixture of water and ice because the setup looks similar to the last problem. We will still start with the all melted scenario just to check. The final temperature is

$$T_f = \frac{m_{\mathrm{w}} c_{\mathrm{w}} T_{\mathrm{w},i} + m_{\mathrm{i}} c_{\mathrm{i}} T_{\mathrm{i},i} - m_{\mathrm{i}} L}{m_{\mathrm{w}} c_{\mathrm{w}} + m_{\mathrm{i}} c_{\mathrm{i}}} = -9.5\ ^\circ\mathrm{C} \tag{27}$$

As expected, this contradicts with the assumption. Now we consider the mixture case. The equation is slightly different from last time because the ice has to first heat up:

$$\Delta Q_{\mathrm{w}} + \Delta Q_{\mathrm{i}} = m_{\mathrm{w}} c_{\mathrm{w}} (T_f - T_{\mathrm{w},i}) + m_{\mathrm{i}} c_{\mathrm{i}} (T_f - T_{\mathrm{i},i}) + m_{\mathrm{i,melted}} L = 0 \tag{28}$$

Using $T_f = 0^\circ$C, the amount of ice melted would be

$$m_{\mathrm{i,melted}} = \frac{m_{\mathrm{w}} c_{\mathrm{w}} (T_{\mathrm{w},i} - T_f) + m_{\mathrm{i}} c_{\mathrm{i}} (T_{\mathrm{i},i} - T_f)}{L} = \frac{20600\ \mathrm{cal}}{80\ \mathrm{cal/g}} = 257.5\ \mathrm{g} \tag{29}$$

There's 1.26 kg of water and 142.5 g of ice remaining.

5. **Measuring specific heat**[6]. To find c_X, the specific heat of material X, I place 75 g of it in a 30 g copper ($c_{\text{copper}} = 0.092$ cal/(g $\cdot$ °C)) calorimeter that contains 65 g of water ($c_{\text{w}} = 1$ cal/(g $\cdot$ °C)), all initially at 20°C. When I add 100g of water at 80°C, the final temperature is 49°C. What is c_X?

   Solution: We know that $T_i \equiv T_{X,i} = T_{\text{copper},i} = T_{\text{w},i} = 20°\text{C}$, and $T_{\text{w,added},i} = 80°\text{C}$. The total heat absorbed by everything should add up to 0:

$$\begin{aligned} 0 &= \Delta Q_X + \Delta Q_{\text{copper}} + \Delta Q_{\text{w}} + \Delta Q_{\text{w,added}} \\ &= m_X c_X (T_f - T_i) + m_{\text{copper}} c_{\text{copper}} (T_f - T_i) \\ &\quad + m_{\text{w}} c_{\text{w}} (T_f - T_i) + m_{\text{w,added}} c_{\text{w}} (T_f - T_{\text{w,added},i}) \end{aligned}$$

   We can rearrange the equation to find

$$\begin{aligned} c_X &= -\frac{1}{m_X} \left(m_{\text{w,added}} c_{\text{w}} \frac{T_f - T_{\text{w,added},i}}{T_f - T_i} + m_{\text{copper}} c_{\text{copper}} + m_{\text{w}} c_{\text{w}} \right) \\ &= 0.52 \text{ cal/(g} \cdot \text{°C)} \end{aligned}$$

[6] Shankar 11.7